Set 3A: Diode Transfer Characteristics

ECEN 325 — Lecture 3 (Diodes) practice set

What this set trains. Finding V_o as a function of V_i for circuits with diodes, resistors and DC references, and sketching it. One method handles every circuit: assume each diode ON or OFF, solve the linear circuit, and keep only the assumptions that check out. Each valid assumption gives one straight piece of the characteristic. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-3 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1/2 tools):

T1 *Diode models.* Ideal: OFF $\Leftrightarrow V_D < 0$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0$). Constant-voltage-drop (CVD): OFF $\Leftrightarrow V_D < 0.7\,\text{V}$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0.7\,\text{V}$ source). Exponential: $I_D = I_S e^{V_D/nV_T}$, i.e. $V_D = nV_T \ln(I_D/I_S)$, $V_T \approx 25\,\text{mV}$.

T2 *State method.* Assume ON/OFF for every diode, solve the now-linear circuit (KCL, Lecture 2), then check the assumptions; each check is a condition on V_i. The valid states tile the V_i axis.

T3 *Small signal.* Around a DC point I_D the diode is a resistor $r_d = \frac{nV_T}{I_D}$ (valid while $|v_d| \ll nV_T$). Capacitors: open at DC; $\frac{1}{sC}$ in AC; DC supplies are AC ground.

T4 *Rectifier with a filter capacitor* ($f = 60\,\text{Hz}$, $T = \frac{1}{f}$ half-wave, $\frac{1}{2f}$ full-wave): peak $V_p = \hat{V}_S -$ (diode drops in the path); ripple $V_r \approx V_p \frac{T}{RC}$; $I_{D,av} \approx I_L\left(1 + \pi\sqrt{2V_p/V_r}\right)$, $I_{D,max} \approx I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right)$; PIV = the largest reverse voltage a diode sees.

Steps:

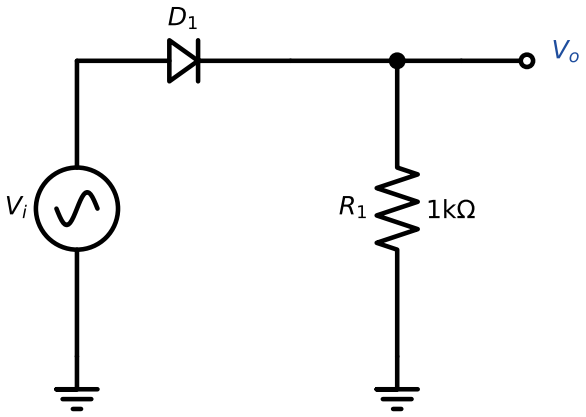
1. *Pick the model* stated in the problem.
2. *Transfer characteristic:* for each ON/OFF state: replace the diodes, solve, check $\Rightarrow$ range of V_i; then join the pieces (they must meet at the breakpoints) and sketch.
3. *Small signal:* DC operating point first, then r_d, then the AC circuit exactly as in Lectures 1–2, then add DC and AC.
4. *Rectifiers:* peak $\rightarrow$ ripple (or C for a given ripple) $\rightarrow$ diode currents $\rightarrow$ PIV $\rightarrow$ transformer ratio.

Problems

Easy problems

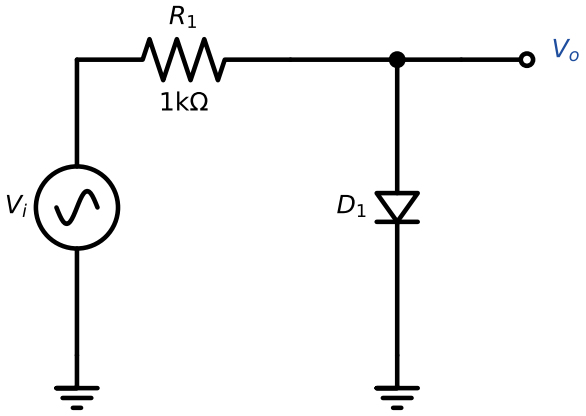
Problem 3A.1 **Easy**

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 5\sin(2\pi \cdot 60\,t)$.



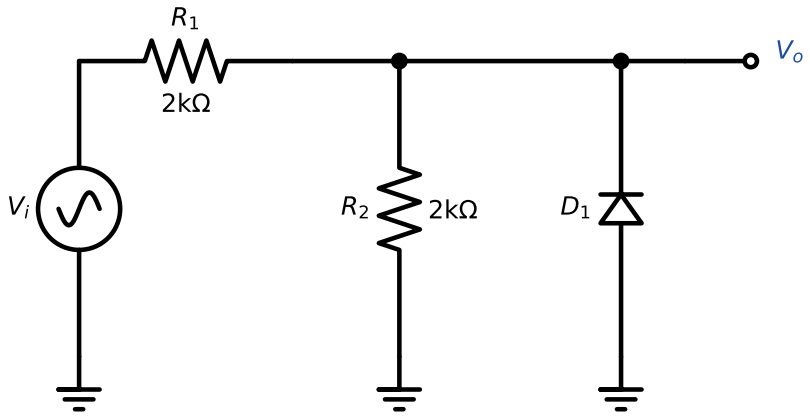
Problem 3A.2 **Easy**

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 3\sin(2\pi \cdot 1000\,t)$.



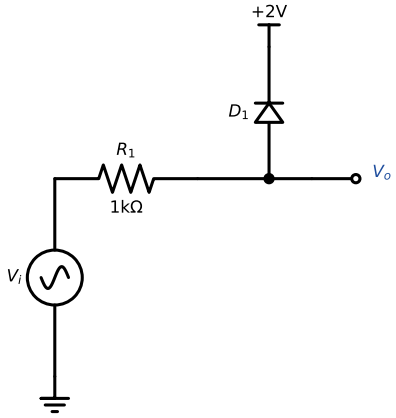
Problem 3A.3 — Easy

For the circuit below, use the ideal-diode model. Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 4\sin(2\pi \cdot 1000\,t)$.



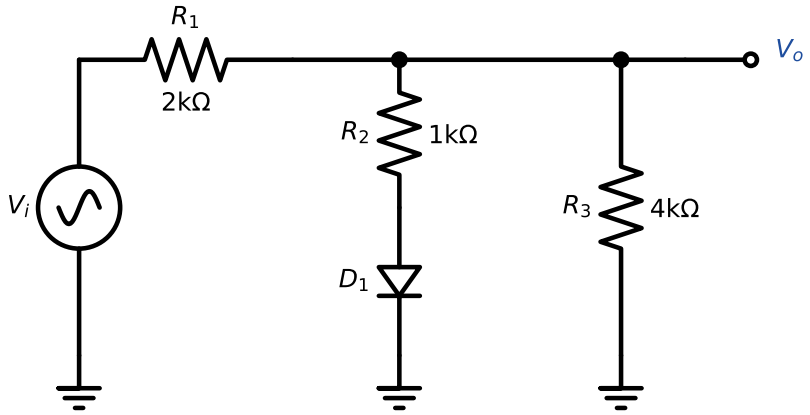
Problem 3A.4 — Easy

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 5\sin(2\pi \cdot 1000\,t)$.



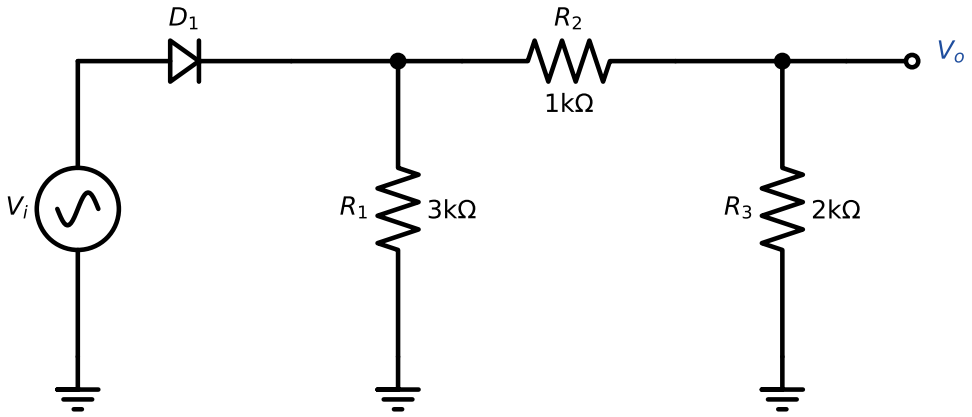
Problem 3A.5 **Easy**

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



Problem 3A.6 **Easy**

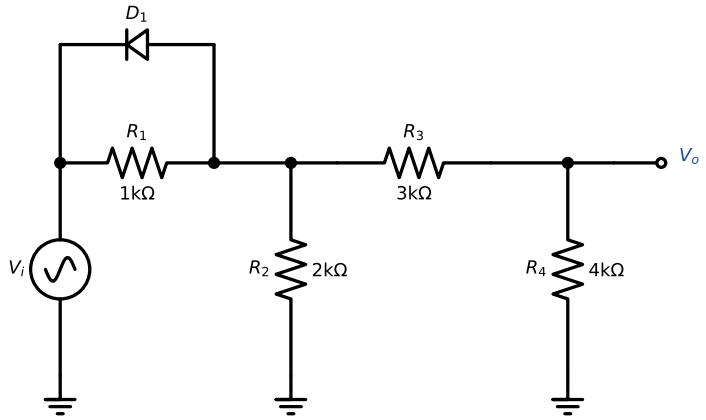
For the circuit below, use the ideal-diode model. Find V_o as a function of V_i and sketch V_o versus V_i.



Medium problems

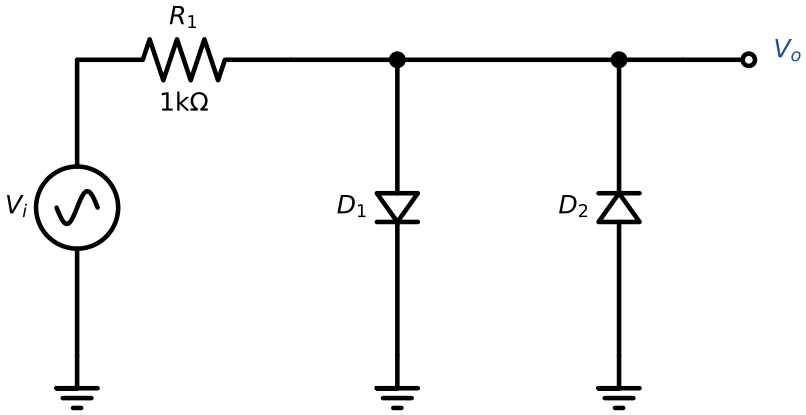
Problem 3A.7 — Medium

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



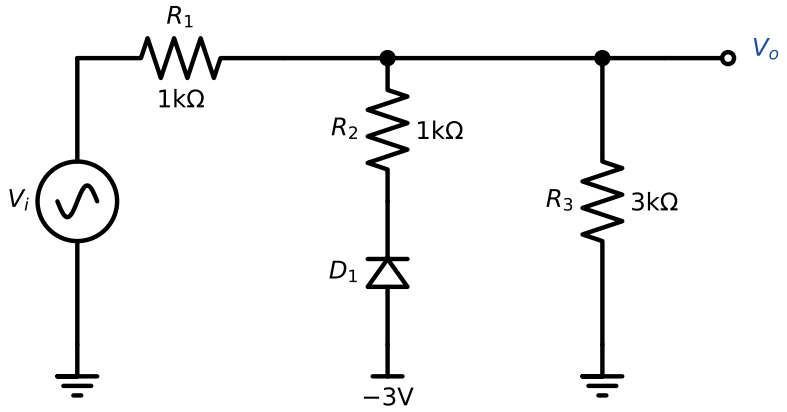
Problem 3A.8 — Medium

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



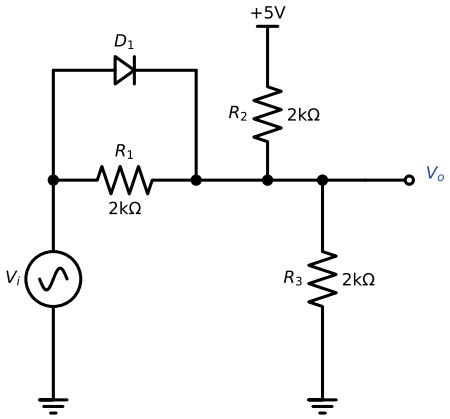
Problem 3A.9 **Medium**

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



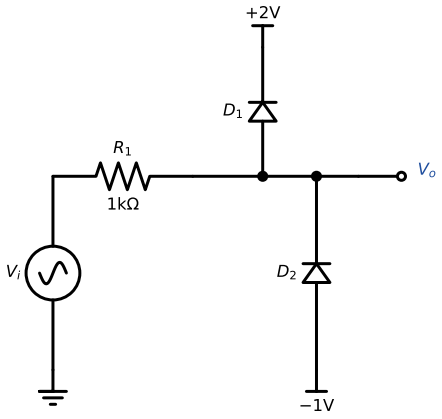
Problem 3A.10 **Medium**

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



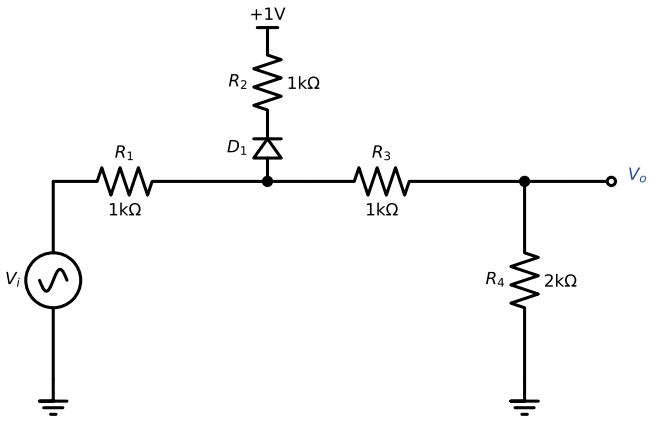
Problem 3A.11 — Medium

For the circuit below, use the ideal-diode model. Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 4\sin(2\pi \cdot 500\,t)$.



Problem 3A.12 — Medium

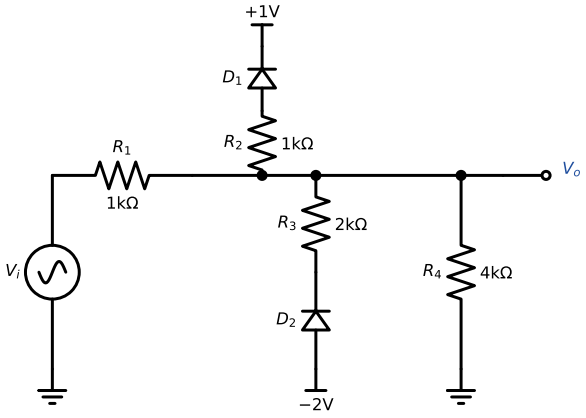
For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



Hard problems

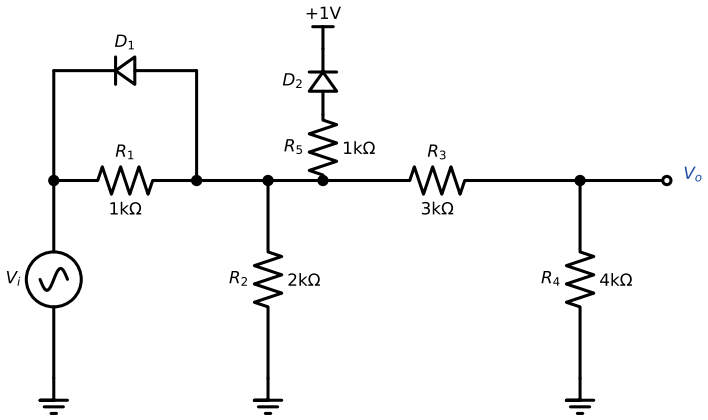
Problem 3A.13 — Hard

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 6\sin(2\pi \cdot 1000\,t)$.



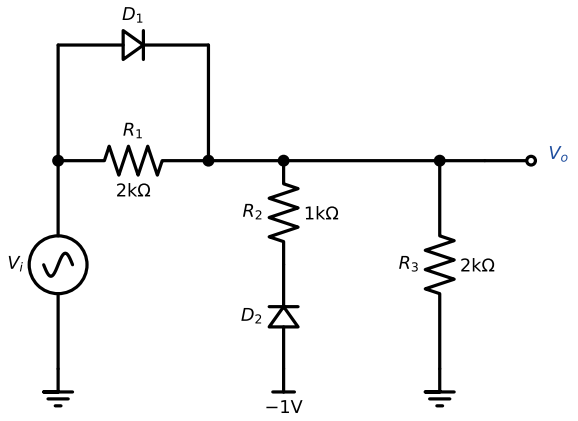
Problem 3A.14 — Hard

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



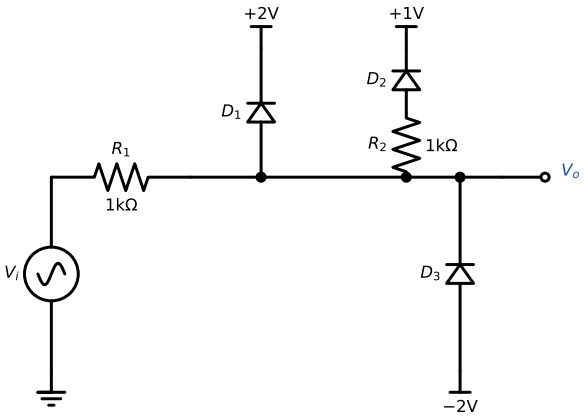
Problem 3A.15 — Hard

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



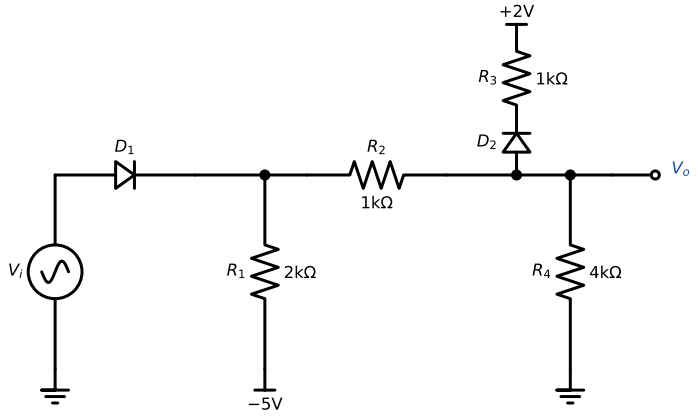
Problem 3A.16 — Hard

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 5\sin(2\pi \cdot 1000\,t)$.



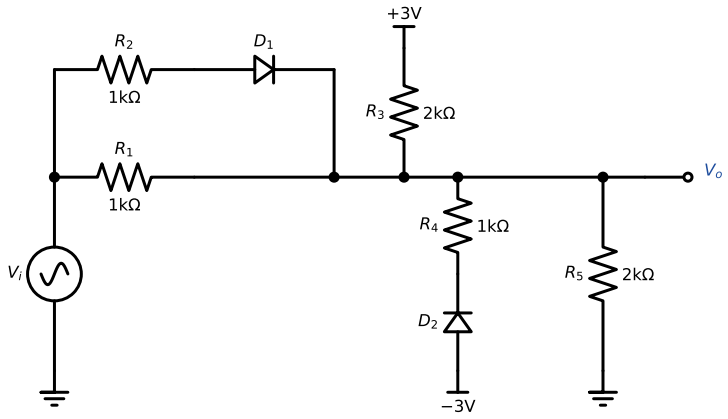
Problem 3A.17 — Hard

For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i.



Problem 3A.18 — Hard

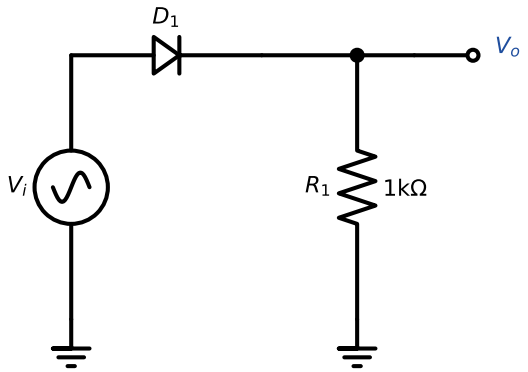
For the circuit below, use the constant-voltage-drop model ($V_{D,on} = 0.7\,\text{V}$). Find V_o as a function of V_i and sketch V_o versus V_i. Also sketch $V_o(t)$ for $V_i(t) = 8\sin(2\pi \cdot 1000\,t)$.



Solutions

Every solution follows the steps of the Lecture-3 pipeline card.

Solution 3A.1 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o}{R_1} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$
$$V_{D_1} = V_i - V_o = V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.7\,\text{V}$. So this state holds for $\boxed{V_i < 0.7}$, where $V_o = 0$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_i - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o}{R_1} - I_{D_1} = 0$$
$$\text{D1 ON}: \qquad V_i - V_o = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i - 0.7$$
$$I_{D_1} = V_i - 0.7\ \text{mA}$$

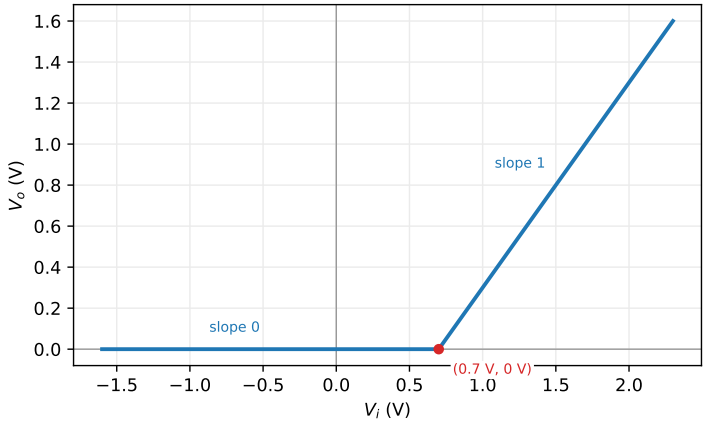
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.7\,\text{V}$. So this state holds for $\boxed{V_i > 0.7}$, where $V_o = V_i - 0.7$.

Step 4 — Assemble and sketch

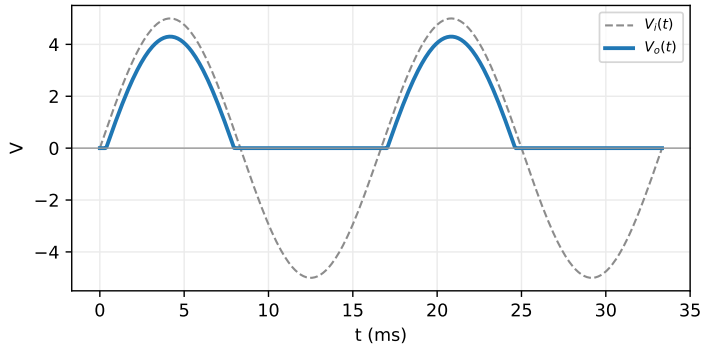
Put the valid states together:

$$V_o = \begin{cases} 0 & V_i < 0.7 \\ V_i - 0.7 & V_i > 0.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0.7\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



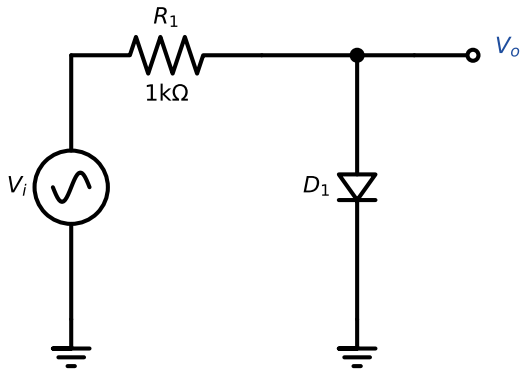
For $V_i(t) = 5\sin(2\pi \cdot 60\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0 & V_i < 0.7 \\ V_i - 0.7 & V_i > 0.7 \end{cases}$$

Solution 3A.2 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i$$
$$V_{D_1} = V_o - 0 = V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.7\,\text{V}$. So this state holds for $\boxed{V_i < 0.7}$, where $V_o = V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_o - 0 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad V_o - 0 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.7$$
$$I_{D_1} = V_i - 0.7\ \text{mA}$$

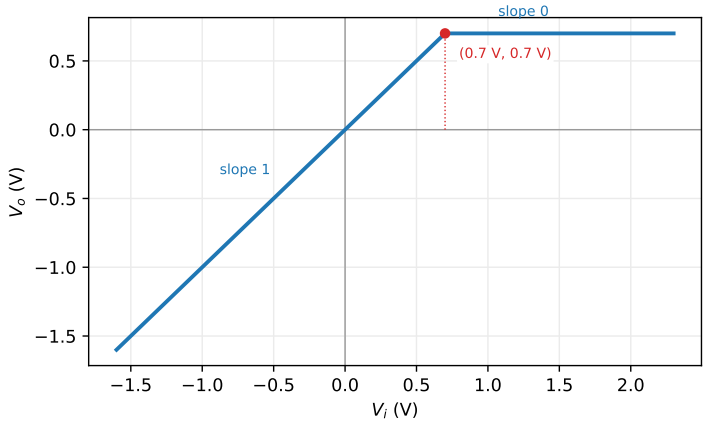
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.7\,\text{V}$. So this state holds for $\boxed{V_i > 0.7}$, where $V_o = 0.7$.

Step 4 — Assemble and sketch

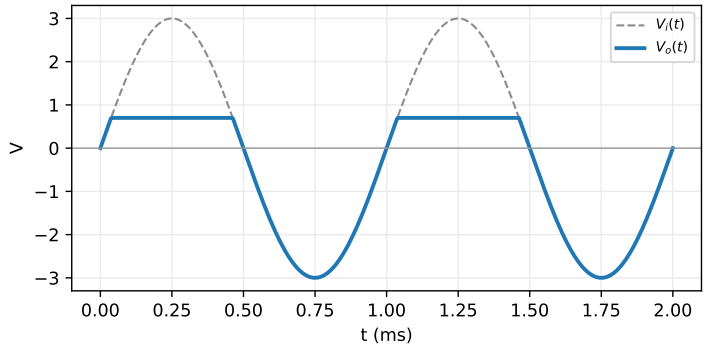
Put the valid states together:

$$V_o = \begin{cases} V_i & V_i < 0.7 \\ 0.7 & V_i > 0.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0.7\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$. ✓



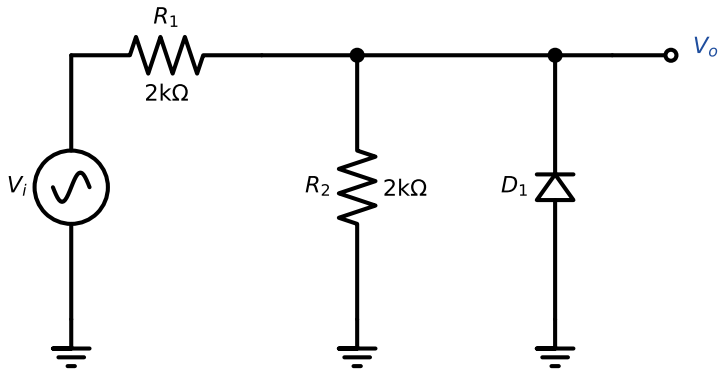
For $V_i(t) = 3\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} V_i & V_i < 0.7 \\ 0.7 & V_i > 0.7 \end{cases}$$

Solution 3A.3 **Easy**



Step 1 — Model and method

Use the ideal-diode model (tool T1): an OFF diode is an open circuit and needs $V_D < 0$; an ON diode is a short ($V_D = 0$) and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 ON

Replace D_1 ON: a 0 V source ($0 - V_o = 0$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + \frac{V_o}{R_2} - I_{D_1} = 0$$
$$\text{D1 ON}: \qquad 0 - V_o = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$
$$I_{D_1} = -0.5V_i \text{ mA}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i < 0$ V. So this state holds for $\boxed{V_i < 0}$, where $V_o = 0$.

Step 3 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + \frac{V_o}{R_2} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.5V_i$$
$$V_{D_1} = 0 - V_o = -0.5V_i$$

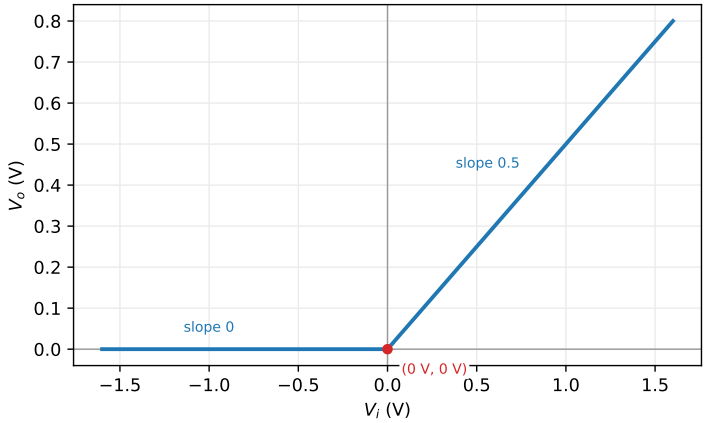
Check the assumptions: $V_{D_1} < 0 \Rightarrow V_i > 0$ V. So this state holds for $\boxed{V_i > 0}$, where $V_o = 0.5V_i$.

Step 4 — Assemble and sketch

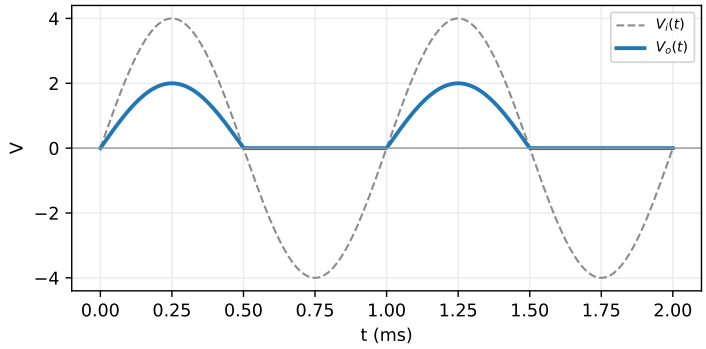
Put the valid states together:

$$V_o = \begin{cases} 0 & V_i < 0 \\ 0.5V_i & V_i > 0 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



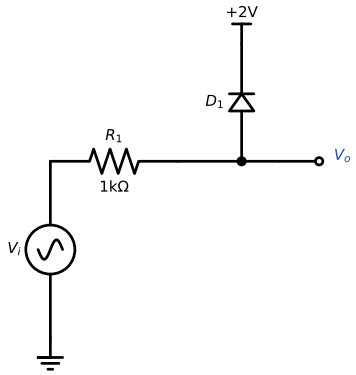
For $V_i(t) = 4\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0 & V_i < 0 \\ 0.5V_i & V_i > 0 \end{cases}$$

Solution 3A.4 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i$$
$$V_{D_1} = V_o - 2 = V_i - 2$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.7\,\text{V}$. So this state holds for $\boxed{V_i < 2.7}$, where $V_o = V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_o - 2 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad V_o - 2 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 2.7$$
$$I_{D_1} = V_i - 2.7\ \text{mA}$$

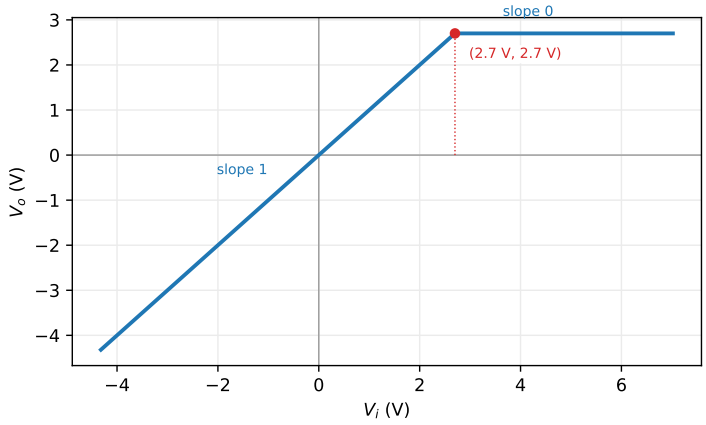
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 2.7\,\text{V}$. So this state holds for $\boxed{V_i > 2.7}$, where $V_o = 2.7$.

Step 4 — Assemble and sketch

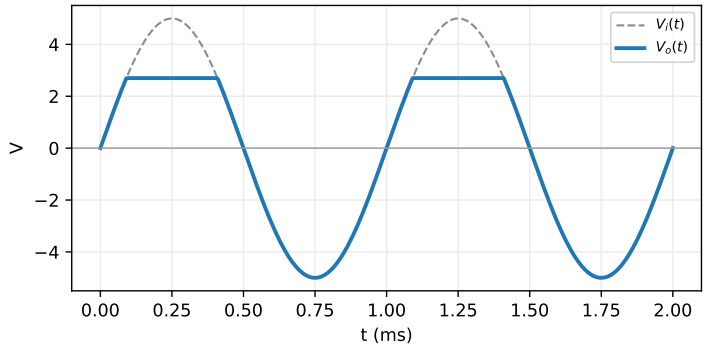
Put the valid states together:

$$V_o = \begin{cases} V_i & V_i < 2.7 \\ 2.7 & V_i > 2.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = 2.7\,\text{V}$ both pieces give $V_o = 2.7\,\text{V}$. ✓



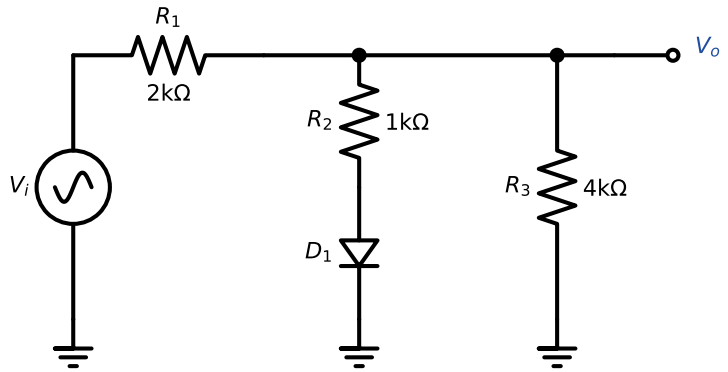
For $V_i(t) = 5\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} V_i & V_i < 2.7 \\ 2.7 & V_i > 2.7 \end{cases}$$

Solution 3A.5 **Easy**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_a = node joining D_1, R_2.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_a: \quad \frac{V_a - V_o}{R_2} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.6667 V_i$$
$$V_{D_1} = V_a - 0 = 0.6667 V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 1.05\,\text{V}$. So this state holds for $\boxed{V_i < 1.05}$, where $V_o = 0.6667 V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_a - 0 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_a: \quad \frac{V_a - V_o}{R_2} + I_{D_1} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} = 0$$

$$\text{D1 ON}: \quad V_a - 0 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.2857 V_i + 0.4$$

$$I_{D_1} = 0.2857V_i - 0.3 \text{ mA}$$

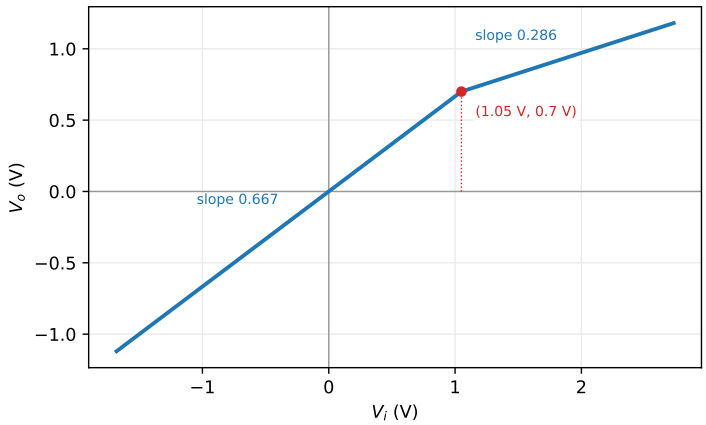
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 1.05\,\text{V}$. So this state holds for $\boxed{V_i > 1.05}$, where $V_o = 0.2857V_i + 0.4$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.6667V_i & V_i < 1.05 \\ 0.2857V_i + 0.4 & V_i > 1.05 \end{cases}$$

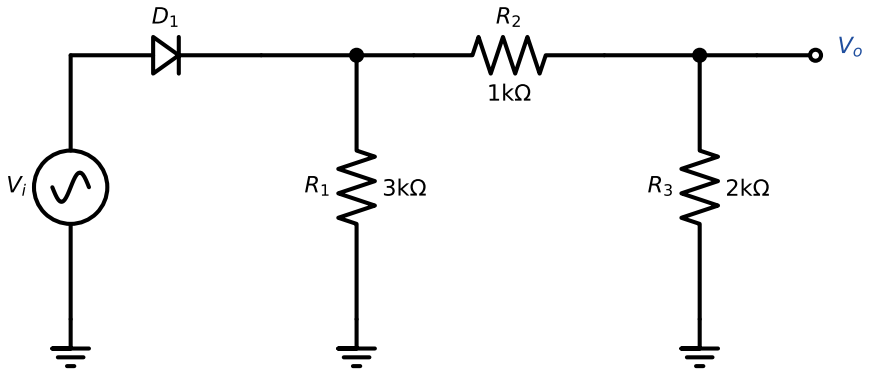
Check: the pieces must meet at each breakpoint — at $V_i = 1.05\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.6667V_i & V_i < 1.05 \\ 0.2857V_i + 0.4 & V_i > 1.05 \end{cases}$$

Solution 3A.6 **Easy**



Step 1 — Model and method

Use the ideal-diode model (tool T1): an OFF diode is an open circuit and needs $V_D < 0$; an ON diode is a short ($V_D = 0$) and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_1 = node joining D_1, R_1, R_2.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \quad \frac{V_1}{R_1} + \frac{V_1 - V_o}{R_2} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_1}{R_2} + \frac{V_o}{R_3} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0$$
$$V_{D_1} = V_i - V_1 = V_i$$

Check the assumptions: $V_{D_1} < 0 \Rightarrow V_i < 0\,\text{V}$. So this state holds for $\boxed{V_i < 0}$, where $V_o = 0$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0 V source ($V_i - V_1 = 0$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \quad \frac{V_1}{R_1} + \frac{V_1 - V_o}{R_2} - I_{D_1} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_1}{R_2} + \frac{V_o}{R_3} = 0$$

$$\text{D1 ON}: \quad V_i - V_1 = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.6667 V_i$$

$$I_{D_1} = 0.6667 V_i \text{ mA}$$

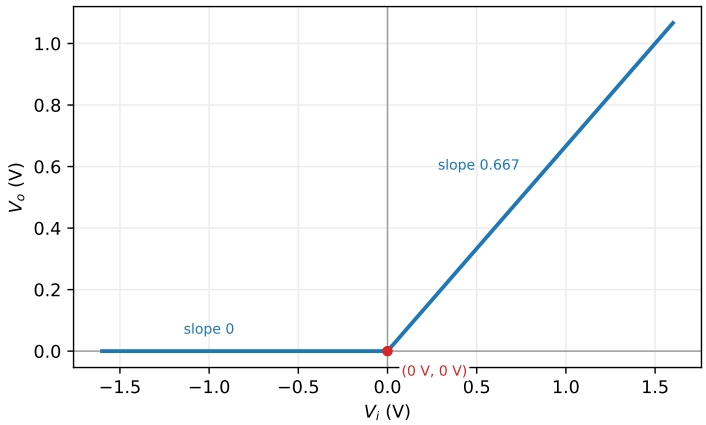
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0\,\text{V}$. So this state holds for $\boxed{V_i > 0}$, where $V_o = 0.6667 V_i$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0 & V_i < 0 \\ 0.6667 V_i & V_i > 0 \end{cases}$$

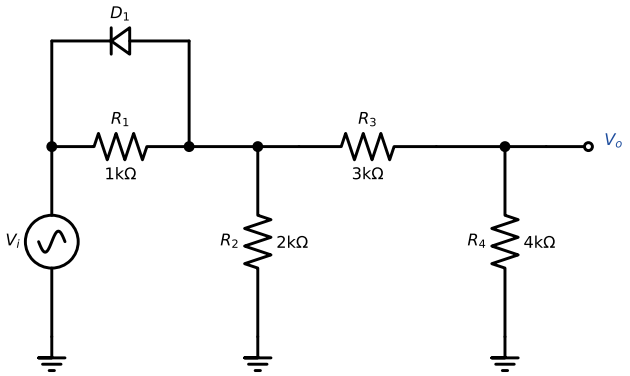
Check: the pieces must meet at each breakpoint — at $V_i = 0\,\text{V}$ both pieces give $V_o = 0\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0 & V_i < 0 \\ 0.6667 V_i & V_i > 0 \end{cases}$$

Solution 3A.7 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_1 = node joining D_1, R_1, R_2, R_3.

Step 2 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_1 - V_i = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_1: && \frac{V_1 - V_i}{R_1} + \frac{V_1}{R_2} + \frac{V_1 - V_o}{R_3} + I_{D_1} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0 \\
&\text{D1 ON}: && V_1 - V_i = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.5714 V_i + 0.4 \\
I_{D_1} &= -0.6429 V_i - 1.15\ \text{mA}
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i < -1.789\,\text{V}$. So this state holds for $\boxed{V_i < -1.789}$, where $V_o = 0.5714 V_i + 0.4$.

Step 3 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_1: && \frac{V_1 - V_i}{R_1} + \frac{V_1}{R_2} + \frac{V_1 - V_o}{R_3} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.3478 V_i$$

$$V_{D_1} = V_1 - V_i = -0.3913 V_i$$

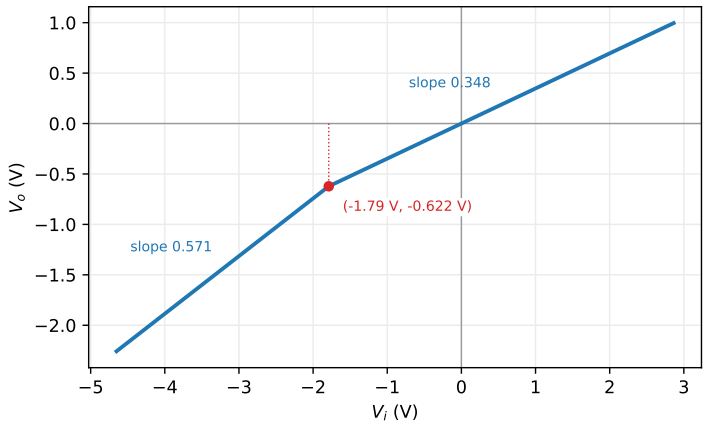
Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i > -1.789\,\text{V}$. So this state holds for $\boxed{V_i > -1.789}$, where $V_o = 0.3478 V_i$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.5714 V_i + 0.4 & V_i < -1.789 \\ 0.3478 V_i & V_i > -1.789 \end{cases}$$

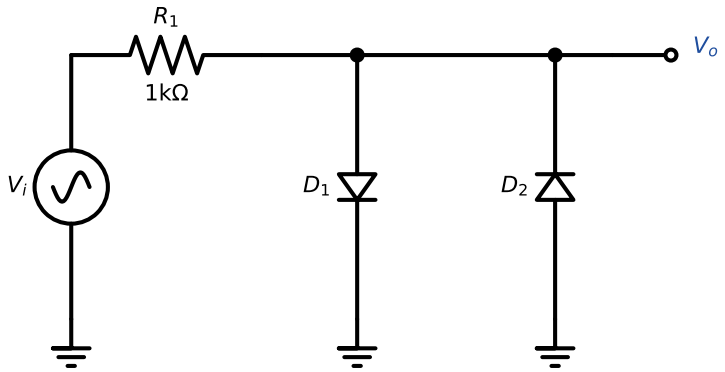
Check: the pieces must meet at each breakpoint — at $V_i = -1.789\,\text{V}$ both pieces give $V_o = -0.6222\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.5714 V_i + 0.4 & V_i < -1.789 \\ 0.3478 V_i & V_i > -1.789 \end{cases}$$

Solution 3A.8 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($0 - V_o = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} - I_{D_2} = 0 \\ &\text{D2 ON}: && 0 - V_o = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -0.7 \\ V_{D_1} = V_o - 0 &= -0.7 \\ I_{D_2} &= -V_i - 0.7 \text{ mA} \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow$ true for every V_i; $I_{D_2} > 0 \Rightarrow V_i < -0.7\,\text{V}$. So this state holds for $\boxed{V_i < -0.7}$, where $V_o = -0.7$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i$$

$$V_{D_1} = V_o - 0 = V_i$$
$$V_{D_2} = 0 - V_o = -V_i$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 0.7\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -0.7\,\text{V}$. So this state holds for $\boxed{-0.7 < V_i < 0.7}$, where $V_o = V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_o - 0 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad V_o - 0 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.7$$
$$I_{D_1} = V_i - 0.7 \text{ mA}$$
$$V_{D_2} = 0 - V_o = -0.7$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 0.7\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 0.7}$, where $V_o = 0.7$.

Step 5 — The other combinations

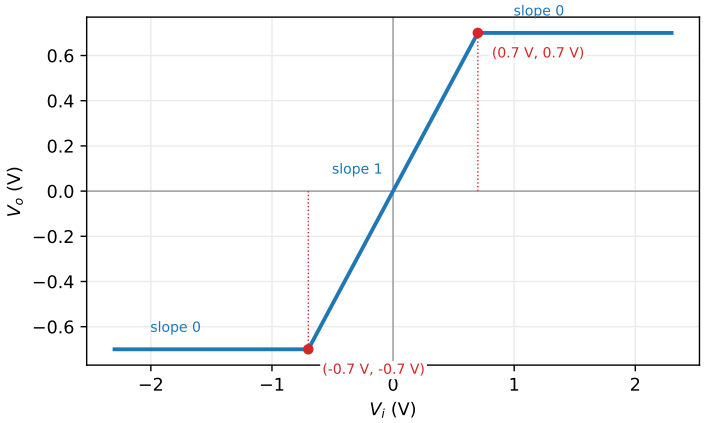
- D_1 ON, D_2 ON: the ON diodes would have to hold the same node at two different voltages at once, which no circuit can do — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

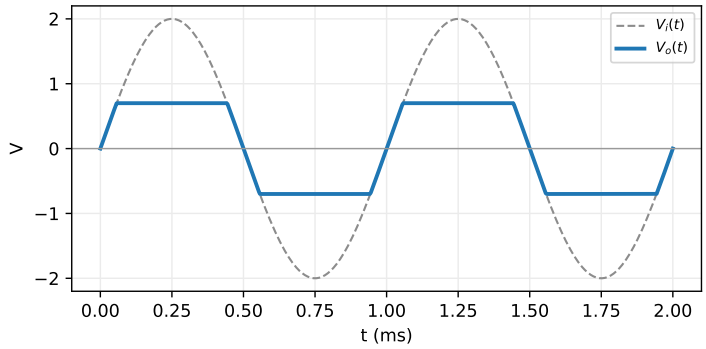
Put the valid states together:

$$V_o = \begin{cases} -0.7 & V_i < -0.7 \\ V_i & -0.7 < V_i < 0.7 \\ 0.7 & V_i > 0.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -0.7\,\text{V}$ both pieces give $V_o = -0.7\,\text{V}$; at $V_i = 0.7\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$. ✓



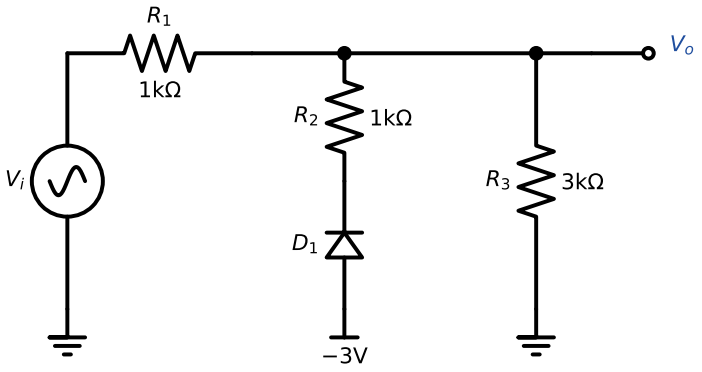
For $V_i(t) = 2\sin(2\pi \cdot 1000\, t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -0.7 & V_i < -0.7 \\ V_i & -0.7 < V_i < 0.7 \\ 0.7 & V_i > 0.7 \end{cases}$$

Solution 3A.9 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_a = node joining D_1, R_2.

Step 2 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($-3 - V_a = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} - I_{D_1} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} = 0 \\
&\text{D1 ON}: && -3 - V_a = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.4286 V_i - 1.586 \\
I_{D_1} &= -0.4286 V_i - 2.114\ \text{mA}
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i < -4.933\,\text{V}$. So this state holds for $\boxed{V_i < -4.933}$, where $V_o = 0.4286 V_i - 1.586$.

Step 3 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} = 0
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.75 V_i$$

$$V_{D_1} = -3 - V_a = -0.75V_i - 3$$

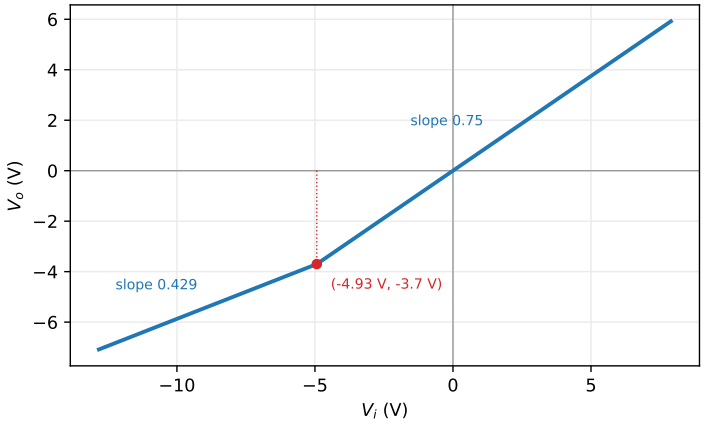
Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i > -4.933\,\text{V}$. So this state holds for $\boxed{V_i > -4.933}$, where $V_o = 0.75V_i$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.4286V_i - 1.586 & V_i < -4.933 \\ 0.75V_i & V_i > -4.933 \end{cases}$$

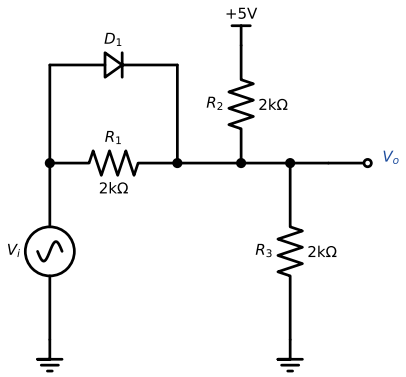
Check: the pieces must meet at each breakpoint — at $V_i = -4.933\,\text{V}$ both pieces give $V_o = -3.7\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.4286V_i - 1.586 & V_i < -4.933 \\ 0.75V_i & V_i > -4.933 \end{cases}$$

Solution 3A.10 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + \frac{V_o - 5}{R_2} + \frac{V_o}{R_3} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.3333V_i + 1.667$$
$$V_{D_1} = V_i - V_o = 0.6667V_i - 1.667$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 3.55\,\text{V}$. So this state holds for $\boxed{V_i < 3.55}$, where $V_o = 0.3333V_i + 1.667$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_i - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + \frac{V_o - 5}{R_2} + \frac{V_o}{R_3} - I_{D_1} = 0$$
$$\text{D1 ON}: \qquad V_i - V_o = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i - 0.7$$
$$I_{D_1} = V_i - 3.55\ \text{mA}$$

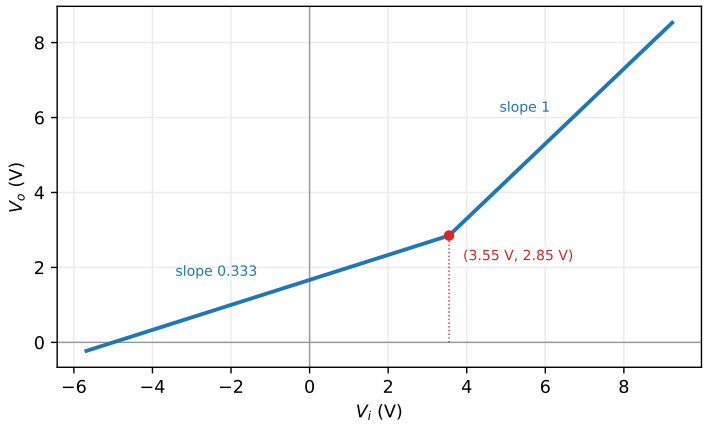
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 3.55\,\text{V}$. So this state holds for $\boxed{V_i > 3.55}$, where $V_o = V_i - 0.7$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.3333 V_i + 1.667 & V_i < 3.55 \\ V_i - 0.7 & V_i > 3.55 \end{cases}$$

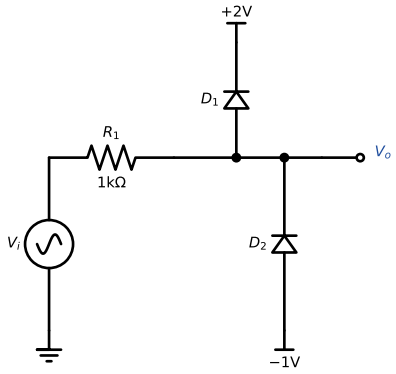
Check: the pieces must meet at each breakpoint — at $V_i = 3.55\,\text{V}$ both pieces give $V_o = 2.85\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.3333 V_i + 1.667 & V_i < 3.55 \\ V_i - 0.7 & V_i > 3.55 \end{cases}$$

Solution 3A.11 **Medium**



Step 1 — Model and method

Use the ideal-diode model (tool T1): an OFF diode is an open circuit and needs $V_D < 0$; an ON diode is a short ($V_D = 0$) and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2).

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0 V source ($-1 - V_o = 0$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} - I_{D_2} = 0 \\ &\text{D2 ON}: && -1 - V_o = 0 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -1 \\ V_{D_1} = V_o - 2 &= -3 \\ I_{D_2} &= -V_i - 1 \text{ mA} \end{aligned}$$

Check the assumptions: $V_{D_1} < 0 \Rightarrow$ true for every V_i; $I_{D_2} > 0 \Rightarrow V_i < -1$ V. So this state holds for $\boxed{V_i < -1}$, where $V_o = -1$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = V_i$$

$$V_{D_1} = V_o - 2 = V_i - 2$$
$$V_{D_2} = -1 - V_o = -V_i - 1$$

Check the assumptions: $V_{D_1} < 0 \Rightarrow V_i < 2\,\text{V}$; $V_{D_2} < 0 \Rightarrow V_i > -1\,\text{V}$. So this state holds for $\boxed{-1 < V_i < 2}$, where $V_o = V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0 V source ($V_o - 2 = 0$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + I_{D_1} = 0$$
$$\text{D1 ON}: \qquad V_o - 2 = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 2$$
$$I_{D_1} = V_i - 2 \text{ mA}$$
$$V_{D_2} = -1 - V_o = -3$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 2\,\text{V}$; $V_{D_2} < 0 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 2}$, where $V_o = 2$.

Step 5 — The other combinations

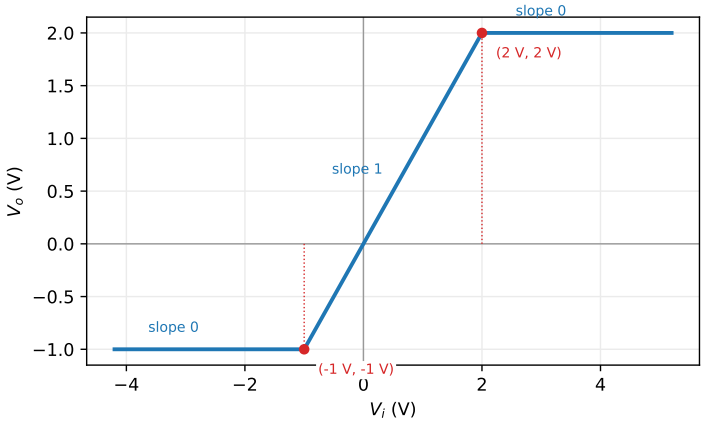
- D_1 ON, D_2 ON: the ON diodes would have to hold the same node at two different voltages at once, which no circuit can do — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

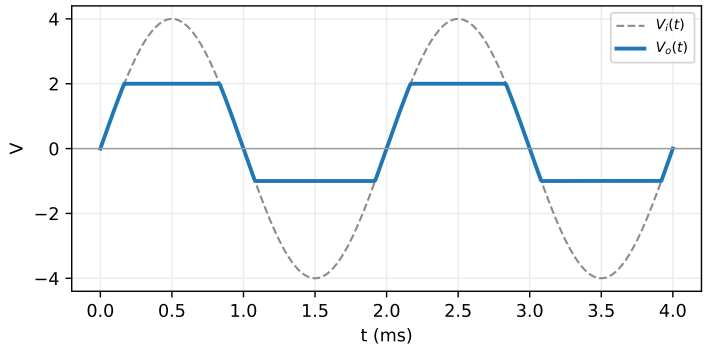
Put the valid states together:

$$V_o = \begin{cases} -1 & V_i < -1 \\ V_i & -1 < V_i < 2 \\ 2 & V_i > 2 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -1\,\text{V}$ both pieces give $V_o = -1\,\text{V}$; at $V_i = 2\,\text{V}$ both pieces give $V_o = 2\,\text{V}$. ✓



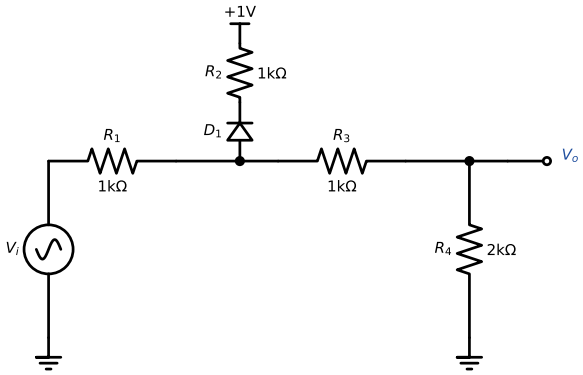
For $V_i(t) = 4\sin(2\pi \cdot 500\, t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -1 & V_i < -1 \\ V_i & -1 < V_i < 2 \\ 2 & V_i > 2 \end{cases}$$

Solution 3A.12 **Medium**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 1 diode there are 2 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_1 = node joining D_1, R_1, R_3; V_a = node joining D_1, R_2.

Step 2 — Assume D_1 OFF

Replace D_1 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \quad \frac{V_1 - V_i}{R_1} + \frac{V_1 - V_o}{R_3} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a - 1}{R_2} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.5V_i$$
$$V_{D_1} = V_1 - V_a = 0.75V_i - 1$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.267\,\text{V}$. So this state holds for $\boxed{V_i < 2.267}$, where $V_o = 0.5V_i$.

Step 3 — Assume D_1 ON

Replace D_1 ON: a 0.7 V source ($V_1 - V_a = 0.7$) carrying an unknown current I_{D_1} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \quad \frac{V_1 - V_i}{R_1} + \frac{V_1 - V_o}{R_3} + I_{D_1} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a - 1}{R_2} - I_{D_1} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0$$

D1 ON : $V_1 - V_a = 0.7$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.2857V_i + 0.4857$$
$$I_{D_1} = 0.4286V_i - 0.9714 \text{ mA}$$

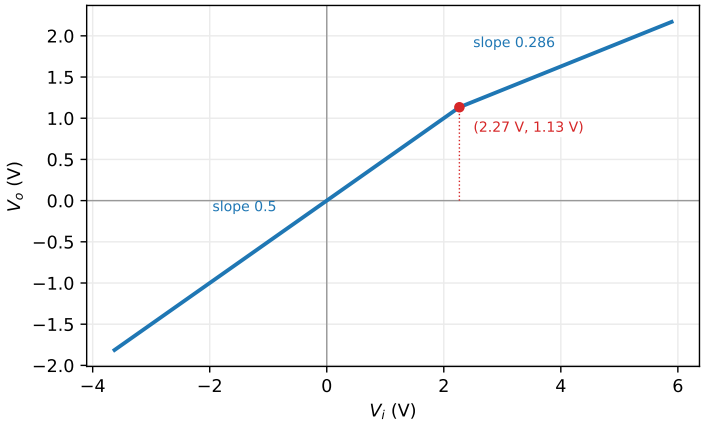
Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 2.267\,\text{V}$. So this state holds for $\boxed{V_i > 2.267}$, where $V_o = 0.2857V_i + 0.4857$.

Step 4 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.5V_i & V_i < 2.267 \\ 0.2857V_i + 0.4857 & V_i > 2.267 \end{cases}$$

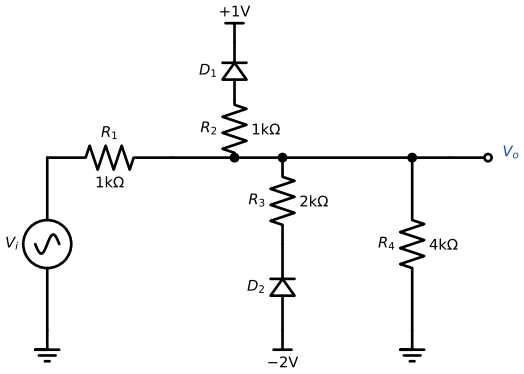
Check: the pieces must meet at each breakpoint — at $V_i = 2.267\,\text{V}$ both pieces give $V_o = 1.133\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.5V_i & V_i < 2.267 \\ 0.2857V_i + 0.4857 & V_i > 2.267 \end{cases}$$

Solution 3A.13 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_a = node joining D_1, R_2; V_b = node joining D_2, R_3.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($-2 - V_b = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} = 0\\
&\text{KCL at } V_b: && \frac{V_b - V_o}{R_3} - I_{D_2} = 0\\
&\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o - V_b}{R_3} + \frac{V_o}{R_4} = 0\\
&\text{D2 ON}: && -2 - V_b = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.5714V_i - 0.7714\\
V_{D_1} = V_a - 1 &= 0.5714V_i - 1.771\\
I_{D_2} &= -0.2857V_i - 0.9643\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 4.325\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < -3.375\,\text{V}$. So this state holds for $\boxed{V_i < -3.375}$, where $V_o = 0.5714V_i - 0.7714$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_a: \quad \frac{V_a - V_o}{R_2} = 0$$

$$\text{KCL at } V_b: \quad \frac{V_b - V_o}{R_3} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o - V_b}{R_3} + \frac{V_o}{R_4} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.8V_i$$
$$V_{D_1} = V_a - 1 = 0.8V_i - 1$$
$$V_{D_2} = -2 - V_b = -0.8V_i - 2$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.125\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -3.375\,\text{V}$. So this state holds for $\boxed{-3.375 < V_i < 2.125}$, where $V_o = 0.8V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_a - 1 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_a: \quad \frac{V_a - V_o}{R_2} + I_{D_1} = 0$$

$$\text{KCL at } V_b: \quad \frac{V_b - V_o}{R_3} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o - V_b}{R_3} + \frac{V_o}{R_4} = 0$$

$$\text{D1 ON}: \quad V_a - 1 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.4444V_i + 0.7556$$
$$I_{D_1} = 0.4444V_i - 0.9444\ \text{mA}$$
$$V_{D_2} = -2 - V_b = -0.4444V_i - 2.756$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 2.125\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -7.775\,\text{V}$. So this state holds for $\boxed{V_i > 2.125}$, where $V_o = 0.4444V_i + 0.7556$.

Step 5 — The other combinations

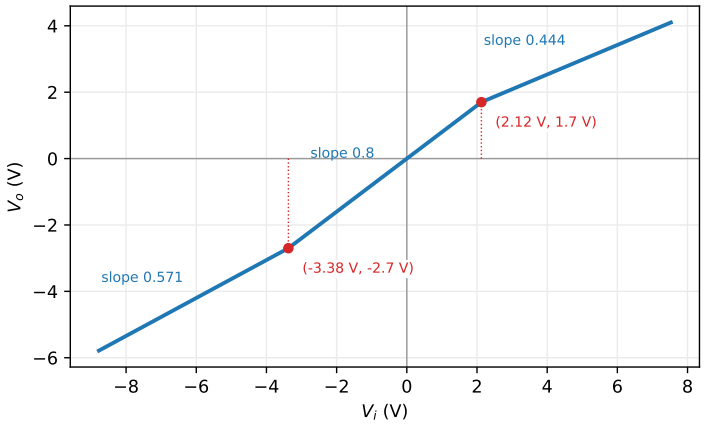
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > 4.325$; $I_{D_2} > 0$ needs $V_i < -7.775$ — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

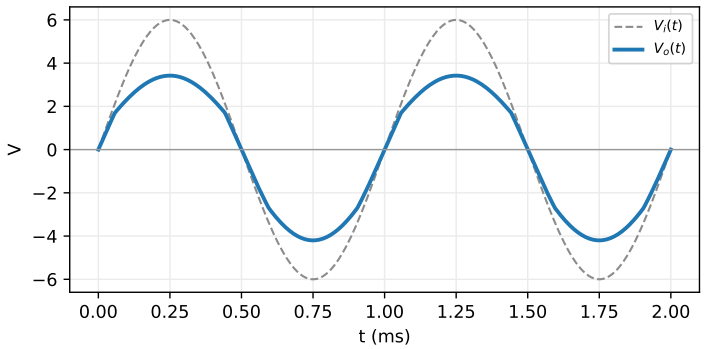
Put the valid states together:

$$V_o = \begin{cases} 0.5714V_i - 0.7714 & V_i < -3.375 \\ 0.8V_i & -3.375 < V_i < 2.125 \\ 0.4444V_i + 0.7556 & V_i > 2.125 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -3.375\,\text{V}$ both pieces give $V_o = -2.7\,\text{V}$; at $V_i = 2.125\,\text{V}$ both pieces give $V_o = 1.7\,\text{V}$. ✓



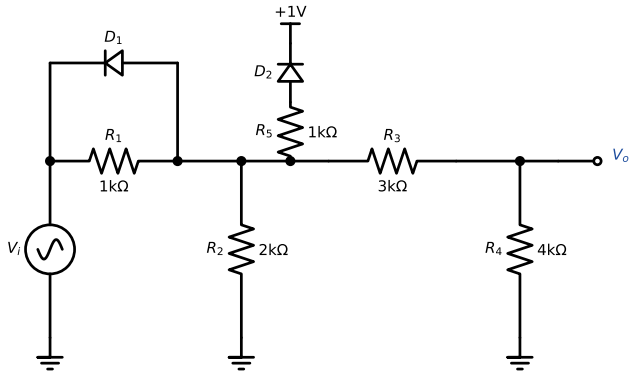
For $V_i(t) = 6\sin(2\pi \cdot 1000\, t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0.5714V_i - 0.7714 & V_i < -3.375 \\ 0.8V_i & -3.375 < V_i < 2.125 \\ 0.4444V_i + 0.7556 & V_i > 2.125 \end{cases}$$

Solution 3A.14 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_1 = node joining D_1, R_1, R_2, R_3, R_5; V_a = node joining D_2, R_5.

Step 2 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_1 - V_i = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_1: && \frac{V_1 - V_i}{R_1} + \frac{V_1}{R_2} + \frac{V_1 - V_a}{R_5} + \frac{V_1 - V_o}{R_3} + I_{D_1} = 0 \\
&\text{KCL at } V_a: && \frac{V_a - V_1}{R_5} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0 \\
&\text{D1 ON}: && V_1 - V_i = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.5714 V_i + 0.4 \\
I_{D_1} &= -0.6429 V_i - 1.15\ \text{mA} \\
V_{D_2} = V_a - 1 &= V_i - 0.3
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i < -1.789\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i < 1\,\text{V}$. So this state holds for $\boxed{V_i < -1.789}$, where $V_o = 0.5714 V_i + 0.4$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \qquad \frac{V_1 - V_i}{R_1} + \frac{V_1}{R_2} + \frac{V_1 - V_a}{R_5} + \frac{V_1 - V_o}{R_3} = 0$$

$$\text{KCL at } V_a: \qquad \frac{V_a - V_1}{R_5} = 0$$

$$\text{KCL at } V_o: \qquad \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.3478V_i$$
$$V_{D_1} = V_1 - V_i = -0.3913V_i$$
$$V_{D_2} = V_a - 1 = 0.6087V_i - 1$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i > -1.789\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i < 2.793\,\text{V}$. So this state holds for $\boxed{-1.789 < V_i < 2.793}$, where $V_o = 0.3478V_i$.

Step 4 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_a - 1 = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \qquad \frac{V_1 - V_i}{R_1} + \frac{V_1}{R_2} + \frac{V_1 - V_a}{R_5} + \frac{V_1 - V_o}{R_3} = 0$$

$$\text{KCL at } V_a: \qquad \frac{V_a - V_1}{R_5} + I_{D_2} = 0$$

$$\text{KCL at } V_o: \qquad \frac{V_o - V_1}{R_3} + \frac{V_o}{R_4} = 0$$

$$\text{D2 ON}: \qquad V_a - 1 = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.2162V_i + 0.3676$$
$$V_{D_1} = V_1 - V_i = -0.6216V_i + 0.6432$$
$$I_{D_2} = 0.3784V_i - 1.057\ \text{mA}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i > -0.0913\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i > 2.793\,\text{V}$. So this state holds for $\boxed{V_i > 2.793}$, where $V_o = 0.2162V_i + 0.3676$.

Step 5 — The other combinations

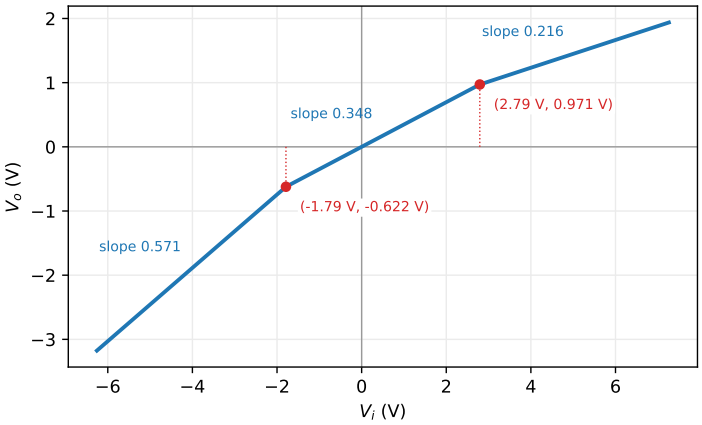
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i < -0.0913$; $I_{D_2} > 0$ needs $V_i > 1$ — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.5714V_i + 0.4 & V_i < -1.789 \\ 0.3478V_i & -1.789 < V_i < 2.793 \\ 0.2162V_i + 0.3676 & V_i > 2.793 \end{cases}$$

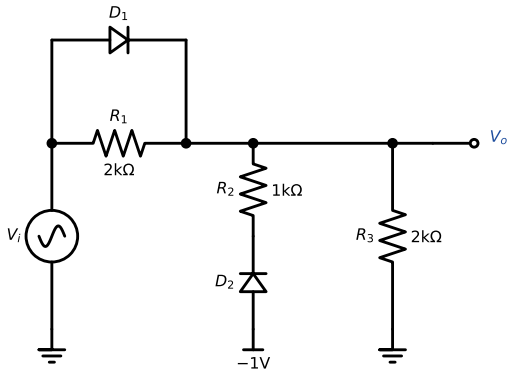
Check: the pieces must meet at each breakpoint — at $V_i = -1.789\,\text{V}$ both pieces give $V_o = -0.6222\,\text{V}$; at $V_i = 2.793\,\text{V}$ both pieces give $V_o = 0.9714\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.5714V_i + 0.4 & V_i < -1.789 \\ 0.3478V_i & -1.789 < V_i < 2.793 \\ 0.2162V_i + 0.3676 & V_i > 2.793 \end{cases}$$

Solution 3A.15 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_a = node joining D_2, R_2.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($-1 - V_a = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} - I_{D_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} = 0 \\ &\text{D2 ON}: && -1 - V_a = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= 0.25V_i - 0.85 \\ V_{D_1} = V_i - V_o &= 0.75V_i + 0.85 \\ I_{D_2} &= -0.25V_i - 0.85 \text{ mA} \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < -0.2\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < -3.4\,\text{V}$. So this state holds for $\boxed{V_i < -3.4}$, where $V_o = 0.25V_i - 0.85$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} = 0 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= 0.5V_i \\ V_{D_1} &= V_i - V_o = 0.5V_i \\ V_{D_2} &= -1 - V_a = -0.5V_i - 1 \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 1.4\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -3.4\,\text{V}$. So this state holds for $\boxed{-3.4 < V_i < 1.4}$, where $V_o = 0.5V_i$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_i - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + \frac{V_o}{R_3} - I_{D_1} = 0 \\ &\text{D1 ON}: && V_i - V_o = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= V_i - 0.7 \\ I_{D_1} &= 0.5V_i - 0.7\ \text{mA} \\ V_{D_2} &= -1 - V_a = -V_i - 0.3 \end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 1.4\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -1\,\text{V}$. So this state holds for $\boxed{V_i > 1.4}$, where $V_o = V_i - 0.7$.

Step 5 — The other combinations

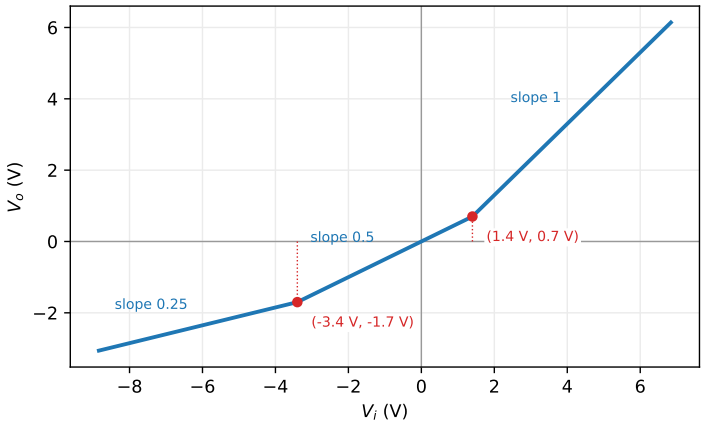
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > -0.2$; $I_{D_2} > 0$ needs $V_i < -1$ — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} 0.25V_i - 0.85 & V_i < -3.4 \\ 0.5V_i & -3.4 < V_i < 1.4 \\ V_i - 0.7 & V_i > 1.4 \end{cases}$$

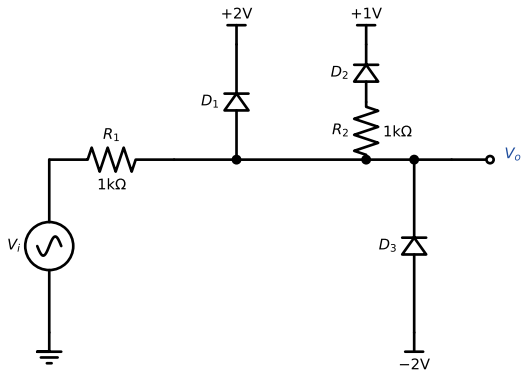
Check: the pieces must meet at each breakpoint — at $V_i = -3.4\,\text{V}$ both pieces give $V_o = -1.7\,\text{V}$; at $V_i = 1.4\,\text{V}$ both pieces give $V_o = 0.7\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} 0.25V_i - 0.85 & V_i < -3.4 \\ 0.5V_i & -3.4 < V_i < 1.4 \\ V_i - 0.7 & V_i > 1.4 \end{cases}$$

Solution 3A.16 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 3 diodes there are 8 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_a = node joining D_2, R_2.

Step 2 — Assume D_1 OFF, D_2 OFF, D_3 ON

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit; D_3 ON: a 0.7 V source ($-2 - V_o = 0.7$) carrying an unknown current I_{D_3} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} - I_{D_3} = 0 \\ &\text{D3 ON}: && -2 - V_o = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= -2.7 \\ V_{D_1} &= V_o - 2 = -4.7 \\ V_{D_2} &= V_a - 1 = -3.7 \\ I_{D_3} &= -V_i - 2.7\ \text{mA} \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow$ true for every V_i; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i; $I_{D_3} > 0 \Rightarrow V_i < -2.7\,\text{V}$. So this state holds for $\boxed{V_i < -2.7}$, where $V_o = -2.7$.

Step 3 — Assume D_1 OFF, D_2 OFF, D_3 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit; D_3 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} = 0 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= V_i \\ V_{D_1} &= V_o - 2 = V_i - 2 \\ V_{D_2} &= V_a - 1 = V_i - 1 \\ V_{D_3} &= -2 - V_o = -V_i - 2 \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.7\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i < 1.7\,\text{V}$; $V_{D_3} < 0.7 \Rightarrow V_i > -2.7\,\text{V}$. So this state holds for $\boxed{-2.7 < V_i < 1.7}$, where $V_o = V_i$.

Step 4 — Assume D_1 OFF, D_2 ON, D_3 OFF

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($V_a - 1 = 0.7$) carrying an unknown current I_{D_2} from anode to cathode; D_3 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} + I_{D_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} = 0 \\ &\text{D2 ON}: && V_a - 1 = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned} V_o &= 0.5V_i + 0.85 \\ V_{D_1} &= V_o - 2 = 0.5V_i - 1.15 \\ I_{D_2} &= 0.5V_i - 0.85\ \text{mA} \\ V_{D_3} &= -2 - V_o = -0.5V_i - 2.85 \end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 3.7\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i > 1.7\,\text{V}$; $V_{D_3} < 0.7 \Rightarrow V_i > -7.1\,\text{V}$. So this state holds for $\boxed{1.7 < V_i < 3.7}$, where $V_o = 0.5V_i + 0.85$.

Step 5 — Assume D_1 ON, D_2 ON, D_3 OFF

Replace D_1 ON: a 0.7 V source ($V_o - 2 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 ON: a 0.7 V source ($V_a - 1 = 0.7$) carrying an unknown current I_{D_2} from anode to cathode; D_3 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned} &\text{KCL at } V_a: && \frac{V_a - V_o}{R_2} + I_{D_2} = 0 \\ &\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - V_a}{R_2} + I_{D_1} = 0 \\ &\text{D1 ON}: && V_o - 2 = 0.7 \\ &\text{D2 ON}: && V_a - 1 = 0.7 \end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 2.7$$

$$I_{D_1} = V_i - 3.7 \text{ mA}$$
$$I_{D_2} = 1 \text{ mA}$$
$$V_{D_3} = -2 - V_o = -4.7$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 3.7\,\text{V}$; $I_{D_2} > 0 \Rightarrow$ true for every V_i; $V_{D_3} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i > 3.7}$, where $V_o = 2.7$.

Step 6 — The other combinations

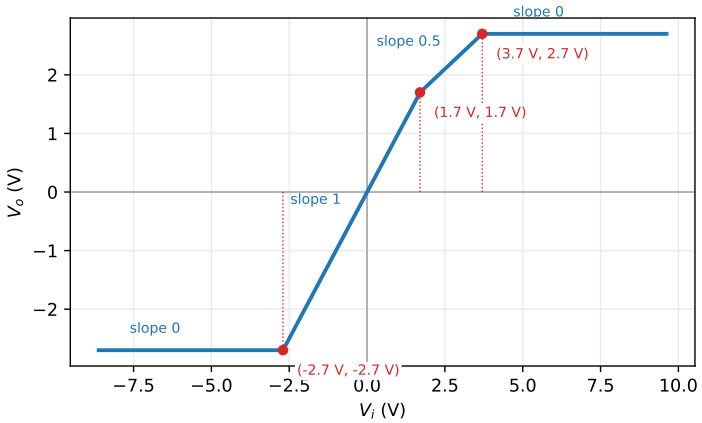
- D_1 OFF, D_2 ON, D_3 ON: $V_{D_1} < 0.7$ is always true; $I_{D_2} > 0$ is never true; $I_{D_3} > 0$ needs $V_i < -7.1$ — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 OFF, D_3 OFF: $I_{D_1} > 0$ needs $V_i > 2.7$; $V_{D_2} < 0.7$ is never true; $V_{D_3} < 0.7$ is always true — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 OFF, D_3 ON: the ON diodes would have to hold the same node at two different voltages at once, which no circuit can do — these cannot all hold, so this state never occurs.
- D_1 ON, D_2 ON, D_3 ON: the ON diodes would have to hold the same node at two different voltages at once, which no circuit can do — these cannot all hold, so this state never occurs.

Step 7 — Assemble and sketch

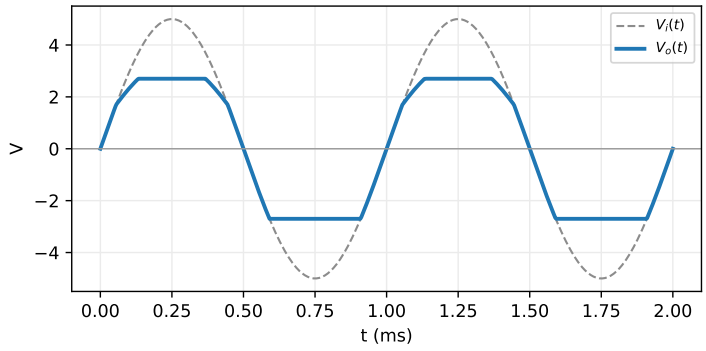
Put the valid states together:

$$V_o = \begin{cases} -2.7 & V_i < -2.7 \\ V_i & -2.7 < V_i < 1.7 \\ 0.5V_i + 0.85 & 1.7 < V_i < 3.7 \\ 2.7 & V_i > 3.7 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -2.7\,\text{V}$ both pieces give $V_o = -2.7\,\text{V}$; at $V_i = 1.7\,\text{V}$ both pieces give $V_o = 1.7\,\text{V}$; at $V_i = 3.7\,\text{V}$ both pieces give $V_o = 2.7\,\text{V}$. ✓



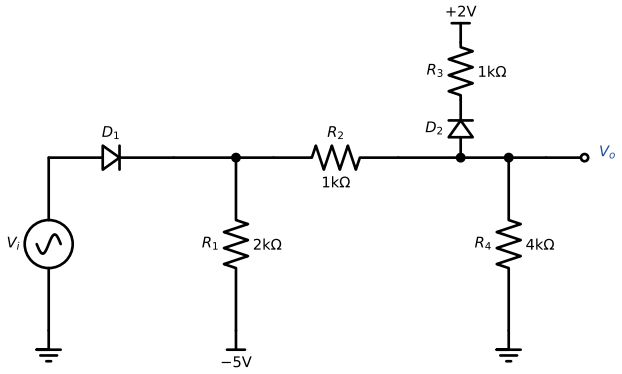
For $V_i(t) = 5\sin(2\pi \cdot 1000\,t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} -2.7 & V_i < -2.7 \\ V_i & -2.7 < V_i < 1.7 \\ 0.5V_i + 0.85 & 1.7 < V_i < 3.7 \\ 2.7 & V_i > 3.7 \end{cases}$$

Solution 3A.17 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_1 = node joining D_1, R_1, R_2; V_a = node joining D_2, R_3.

Step 2 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \quad \frac{V_1 - -5}{R_1} + \frac{V_1 - V_o}{R_2} = 0$$

$$\text{KCL at } V_a: \quad \frac{V_a - 2}{R_3} = 0$$

$$\text{KCL at } V_o: \quad \frac{V_o - V_1}{R_2} + \frac{V_o}{R_4} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = -2.857$$
$$V_{D_1} = V_i - V_1 = V_i + 3.571$$
$$V_{D_2} = V_o - V_a = -4.857$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < -2.871\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow$ true for every V_i. So this state holds for $\boxed{V_i < -2.871}$, where $V_o = -2.857$.

Step 3 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_i - V_1 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_1: \quad \frac{V_1 - -5}{R_1} + \frac{V_1 - V_o}{R_2} - I_{D_1} = 0$$

$$\begin{aligned}
&\text{KCL at } V_a: && \frac{V_a - 2}{R_3} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_1}{R_2} + \frac{V_o}{R_4} = 0 \\
&\text{D1 ON}: && V_i - V_1 = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.8V_i - 0.56 \\
I_{D_1} &= 0.7V_i + 2.01 \text{ mA} \\
V_{D_2} = V_o - V_a &= 0.8V_i - 2.56
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > -2.871\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i < 4.075\,\text{V}$. So this state holds for $\boxed{-2.871 < V_i < 4.075}$, where $V_o = 0.8V_i - 0.56$.

Step 4 — Assume D_1 ON, D_2 ON

Replace D_1 ON: a 0.7 V source ($V_i - V_1 = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 ON: a 0.7 V source ($V_o - V_a = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_1: && \frac{V_1 - -5}{R_1} + \frac{V_1 - V_o}{R_2} - I_{D_1} = 0 \\
&\text{KCL at } V_a: && \frac{V_a - 2}{R_3} - I_{D_2} = 0 \\
&\text{KCL at } V_o: && \frac{V_o - V_1}{R_2} + \frac{V_o}{R_4} + I_{D_2} = 0 \\
&\text{D1 ON}: && V_i - V_1 = 0.7 \\
&\text{D2 ON}: && V_o - V_a = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.4444V_i + 0.8889 \\
I_{D_1} &= 1.056V_i + 0.5611 \text{ mA} \\
I_{D_2} &= 0.4444V_i - 1.811 \text{ mA}
\end{aligned}$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > -0.5316\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i > 4.075\,\text{V}$. So this state holds for $\boxed{V_i > 4.075}$, where $V_o = 0.4444V_i + 0.8889$.

Step 5 — The other combinations

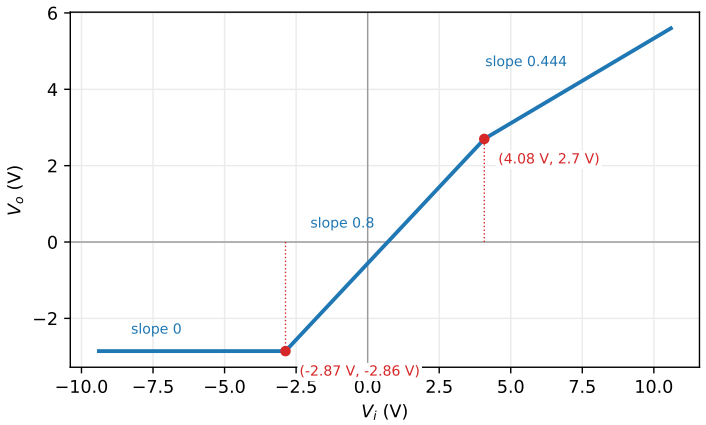
- D_1 OFF, D_2 ON: $V_{D_1} < 0.7$ needs $V_i < -0.5316$; $I_{D_2} > 0$ is never true — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

Put the valid states together:

$$V_o = \begin{cases} -2.857 & V_i < -2.871 \\ 0.8V_i - 0.56 & -2.871 < V_i < 4.075 \\ 0.4444V_i + 0.8889 & V_i > 4.075 \end{cases}$$

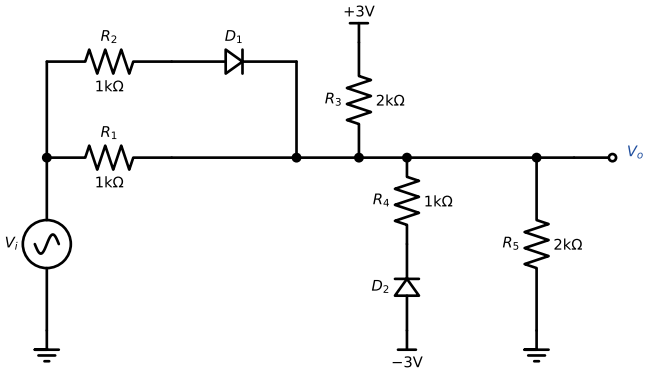
Check: the pieces must meet at each breakpoint — at $V_i = -2.871\,\text{V}$ both pieces give $V_o = -2.857\,\text{V}$; at $V_i = 4.075\,\text{V}$ both pieces give $V_o = 2.7\,\text{V}$. ✓



Answer

$$V_o = \begin{cases} -2.857 & V_i < -2.871 \\ 0.8V_i - 0.56 & -2.871 < V_i < 4.075 \\ 0.4444V_i + 0.8889 & V_i > 4.075 \end{cases}$$

Solution 3A.18 **Hard**



Step 1 — Model and method

Use the constant-voltage-drop model (tool T1): an OFF diode is an open circuit and needs $V_D < 0.7\,\text{V}$; an ON diode is a 0.7 V source and needs $I_D > 0$. With 2 diodes there are 4 ON/OFF combinations. For each one the circuit is linear: solve it with V_i kept as a symbol, then check its assumptions — each check is a condition on V_i (tool T2). Node names used below: V_a = node joining D_1, R_2; V_b = node joining D_2, R_4.

Step 2 — Assume D_1 OFF, D_2 ON

Replace D_1 OFF: an open circuit; D_2 ON: a 0.7 V source ($-3 - V_b = 0.7$) carrying an unknown current I_{D_2} from anode to cathode.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\begin{aligned}
&\text{KCL at } V_a: && \frac{V_a - V_i}{R_2} = 0\\
&\text{KCL at } V_b: && \frac{V_b - V_o}{R_4} - I_{D_2} = 0\\
&\text{KCL at } V_o: && \frac{V_o - V_i}{R_1} + \frac{V_o - 3}{R_3} + \frac{V_o - V_b}{R_4} + \frac{V_o}{R_5} = 0\\
&\text{D2 ON}: && -3 - V_b = 0.7
\end{aligned}$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$\begin{aligned}
V_o &= 0.3333V_i - 0.7333\\
V_{D_1} = V_a - V_o &= 0.6667V_i + 0.7333\\
I_{D_2} &= -0.3333V_i - 2.967\ \text{mA}
\end{aligned}$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < -0.05\,\text{V}$; $I_{D_2} > 0 \Rightarrow V_i < -8.9\,\text{V}$. So this state holds for $\boxed{V_i < -8.9}$, where $V_o = 0.3333V_i - 0.7333$.

Step 3 — Assume D_1 OFF, D_2 OFF

Replace D_1 OFF: an open circuit; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_a: \quad \frac{V_a - V_i}{R_2} = 0$$

$$\text{KCL at } V_b: \qquad \frac{V_b - V_o}{R_4} = 0$$

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + \frac{V_o - 3}{R_3} + \frac{V_o - V_b}{R_4} + \frac{V_o}{R_5} = 0$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.5V_i + 0.75$$
$$V_{D_1} = V_a - V_o = 0.5V_i - 0.75$$
$$V_{D_2} = -3 - V_b = -0.5V_i - 3.75$$

Check the assumptions: $V_{D_1} < 0.7 \Rightarrow V_i < 2.9\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -8.9\,\text{V}$. So this state holds for $\boxed{-8.9 < V_i < 2.9}$, where $V_o = 0.5V_i + 0.75$.

Step 4 — Assume D_1 ON, D_2 OFF

Replace D_1 ON: a 0.7 V source ($V_a - V_o = 0.7$) carrying an unknown current I_{D_1} from anode to cathode; D_2 OFF: an open circuit.

Equations (KCL at every node that is not an op-amp output, plus the diode voltages):

$$\text{KCL at } V_a: \qquad \frac{V_a - V_i}{R_2} + I_{D_1} = 0$$

$$\text{KCL at } V_b: \qquad \frac{V_b - V_o}{R_4} = 0$$

$$\text{KCL at } V_o: \qquad \frac{V_o - V_i}{R_1} + \frac{V_o - 3}{R_3} + \frac{V_o - V_b}{R_4} + \frac{V_o}{R_5} - I_{D_1} = 0$$

$$\text{D1 ON}: \qquad V_a - V_o = 0.7$$

Solving these linear equations (substitute one into the other, as in Lecture 2) gives, with V_i kept as a symbol:

$$V_o = 0.6667V_i + 0.2667$$
$$I_{D_1} = 0.3333V_i - 0.9667\ \text{mA}$$
$$V_{D_2} = -3 - V_b = -0.6667V_i - 3.267$$

Check the assumptions: $I_{D_1} > 0 \Rightarrow V_i > 2.9\,\text{V}$; $V_{D_2} < 0.7 \Rightarrow V_i > -5.95\,\text{V}$. So this state holds for $\boxed{V_i > 2.9}$, where $V_o = 0.6667V_i + 0.2667$.

Step 5 — The other combinations

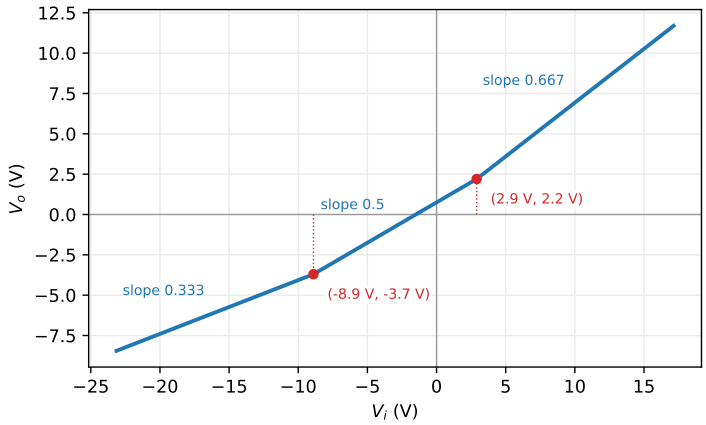
- D_1 ON, D_2 ON: $I_{D_1} > 0$ needs $V_i > -0.05$; $I_{D_2} > 0$ needs $V_i < -5.95$ — these cannot all hold, so this state never occurs.

Step 6 — Assemble and sketch

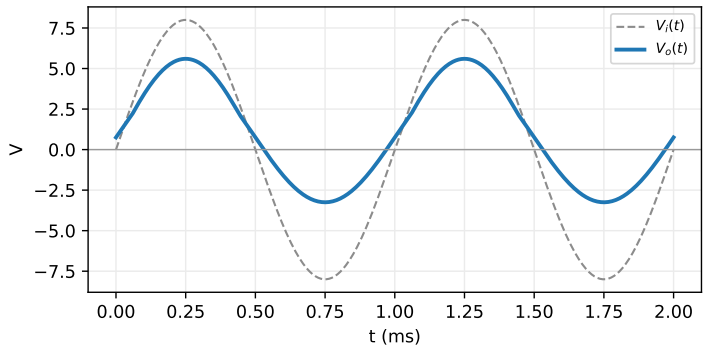
Put the valid states together:

$$V_o = \begin{cases} 0.3333V_i - 0.7333 & V_i < -8.9 \\ 0.5V_i + 0.75 & -8.9 < V_i < 2.9 \\ 0.6667V_i + 0.2667 & V_i > 2.9 \end{cases}$$

Check: the pieces must meet at each breakpoint — at $V_i = -8.9\,\text{V}$ both pieces give $V_o = -3.7\,\text{V}$; at $V_i = 2.9\,\text{V}$ both pieces give $V_o = 2.2\,\text{V}$. ✓



For $V_i(t) = 8\sin(2\pi \cdot 1000\, t)$ simply pass every instant of $V_i(t)$ through the characteristic above (each piece is a straight line, so each part of the sinusoid is scaled and shifted):



Answer

$$V_o = \begin{cases} 0.3333V_i - 0.7333 & V_i < -8.9 \\ 0.5V_i + 0.75 & -8.9 < V_i < 2.9 \\ 0.6667V_i + 0.2667 & V_i > 2.9 \end{cases}$$